

St. Catherine's Medical Center · Dallas, TX 75201

INPATIENT MEDICAL RECORD — CONFIDENTIAL

Patient: Mitchell, Robert T. **MRN:** MRN-2024-001847 **DOB:** 03/15/1966 **Age:** 58 **Sex:** Male
Insurance: BlueCross BlueShield Texas **Subscriber ID:** BCB-7X-4429183
Admission: 01/08/2024 14:32 **Discharge:** 01/13/2024 10:15 **LOS:** 5 Days
Attending: Dr. Sarah K. Patel, MD (Internal Medicine) **Consultant:** Dr. James R. Okonkwo, MD (Pulmonology)

Admitting Diagnosis: COVID-19 with Acute Hypoxic Respiratory Failure

Discharge Diagnoses

	ICD-10	Description
Principal	U07.1	COVID-19
Secondary	J96.01	Acute Respiratory Failure with Hypoxia
Secondary	E11.65	Type 2 Diabetes Mellitus with Hyperglycemia
Secondary	I10	Essential Hypertension
Secondary	E78.5	Hyperlipidemia

DRG: 178 — Respiratory Infections & Inflammations with CC **Billed:** \$24,850.00

ADMISSION HISTORY & PHYSICAL

Date: 01/08/2024 **Time:** 15:47 **Author:** Dr. Sarah K. Patel, MD

Chief Complaint

"I've had a cough and fever for 5 days and I can't catch my breath."

History of Present Illness

Mr. Mitchell is a 58-year-old male with Type 2 DM and Hypertension presenting with 5 days of progressive dyspnea, productive cough, fever, and fatigue. Onset January 3 with home temperature 101.2°F. Cough became productive of yellow-green sputum by January 5. Dyspnea increased to the point of inability to complete sentences without pausing. Seen at urgent care January 7 — COVID-19 rapid antigen result equivocal. Denies chest pain, hemoptysis, or leg swelling. Not vaccinated against COVID-19 in the past 12 months. Works in a large office building.

Past Medical History

- Type 2 Diabetes Mellitus — Metformin 1000 mg BID, Jardiance 10 mg daily

2. Essential Hypertension — Lisinopril 10 mg daily
3. Hyperlipidemia — Atorvastatin 40 mg nightly

Surgical History: Appendectomy (2009); Right knee arthroscopy (2018)

Home Medications

- Metformin 1000 mg PO BID
- Jardiance (Empagliflozin) 10 mg PO daily (*HELD on admission — respiratory illness*)
- Lisinopril 10 mg PO daily
- Atorvastatin 40 mg PO nightly

Allergies: No Known Drug Allergies (NKDA)

Social History: Married. Financial analyst. Former smoker — quit 2014, 10 pack-year history. Occasional alcohol (1–2 drinks/week).

Review of Systems

- *Positive:* Fever, dyspnea, productive cough (yellow-green sputum), fatigue, myalgias, decreased appetite
- *Negative:* Chest pain, hemoptysis, nausea/vomiting, diarrhea, confusion, lower extremity edema

Physical Examination

Parameter	Value	Interpretation
General	Alert, oriented ×3	Mild-moderate respiratory distress
Temperature	38.9°C (102.0°F)	Febrile
Heart Rate	104 bpm	Tachycardic
Respiratory Rate	26 breaths/min	Elevated
Blood Pressure	148/88 mmHg	Mildly elevated
SpO₂ (Room Air)	88%	Below threshold
SpO ₂ after 4L NC	94%	After O ₂ applied
Weight	92 kg, BMI 29.8	

- **HEENT:** Dry mucous membranes. Mild pharyngeal erythema.
- **Respiratory:** Tachypneic. Decreased BS bilateral bases. Bilateral crackles. Accessory muscle use.
- **Cardiovascular:** Tachycardic, regular rhythm. No murmurs/rubs/gallops. No JVD.
- **Abdomen:** Soft, non-tender. No hepatosplenomegaly.
- **Extremities:** No edema. Calves non-tender.

Assessment & Plan

1. **Acute Hypoxic Respiratory Failure / COVID-19 Pneumonia** — Admit to Medical floor with telemetry. COVID-19 PCR STAT. 4L NC, target SpO₂ 92–95%. Dexamethasone 6 mg IV daily. Remdesivir 200 mg IV Day 1, then 100 mg IV Days 2–5. Enoxaparin 40 mg SQ daily. Incentive spirometry. Escalate to HFNC if SpO₂ <90% on 4L NC.
 2. **Type 2 DM** — Hold Jardiance. Slide-scale insulin. Glucose Q6h.
 3. **Hypertension** — Continue Lisinopril 10 mg daily.
 4. **Hyperlipidemia** — Continue Atorvastatin 40 mg nightly.
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LABORATORY RESULTS

Initial Labs — 01/08/2024, 15:55

COVID-19 NAAT (PCR) — Nasopharyngeal Swab

Field	Value
Result	POSITIVE for SARS-CoV-2 RNA
Method	RT-PCR (Abbott Alinity m SARS-CoV-2 Assay)
Resulted	01/08/2024 19:22

Complete Blood Count

Test	Result	Reference	Flag
WBC	11.8 K/uL	4.5–11.0	↑ HIGH
Neutrophils	84%	54–62%	↑ HIGH
Lymphocytes	9%	25–33%	↓ LOW (lymphopenia)
Hemoglobin	14.2 g/dL	13.5–17.5	WNL
Platelets	189 K/uL	150–400	WNL

Comprehensive Metabolic Panel

Test	Result	Reference	Flag
Sodium	138 mEq/L	136–145	WNL
Creatinine	1.1 mg/dL	0.6–1.2	WNL
eGFR	74 mL/min	>60	WNL
Glucose	186 mg/dL	70–99	↑ HIGH
ALT	42 U/L	7–40	↑ Mild

Inflammatory Markers

Test	Result	Reference	Flag
CRP	84.2 mg/L	<10.0	↑ ELEVATED
Ferritin	642 ng/mL	<300	↑ ELEVATED
D-Dimer	0.92 mcg/mL	<0.50	↑ ELEVATED
LDH	312 U/L	122–222	↑ ELEVATED
Procalcitonin	0.18 ng/mL	<0.25	WNL

Arterial Blood Gas — Room Air

Parameter	Value	Reference	Flag
pH	7.44	7.35–7.45	WNL
PaO ₂	54 mmHg	75–100	↓ LOW
PaCO ₂	38 mmHg	35–45	WNL
SaO ₂	86%	94–100%	↓ LOW
P/F Ratio	257	—	Moderate Hypoxemia

Blood Cultures ×2: No growth at 48 hours

Follow-up Labs — 01/10/2024

Test	Result	Trend
CRP	61.4 mg/L	↓ Improving
WBC	9.6 K/uL	↓ Normalizing

Follow-up Labs — 01/12/2024

Test	Result	Trend
CRP	22.1 mg/L	↓ Improving
WBC	8.2 K/uL	Normal
Creatinine	1.0 mg/dL	Stable

RADIOLOGY

Chest X-Ray (PA/Lateral) — 01/08/2024 16:15 (*Dr. Michael Torres, Radiology*)

Impression: Bilateral peripheral and basal patchy opacifications. Right lower lobe consolidation with small pleural effusion. No pneumothorax. Pattern compatible with COVID-19 pneumonia.

CT Thorax without Contrast — 01/08/2024 18:30 (*Dr. Michael Torres, Radiology*)

Impression: Extensive bilateral GGOs with crazy-paving pattern and right lower lobe consolidation involving approximately **40–50% of bilateral lung parenchyma**. Characteristic of COVID-19 pneumonia (CO-RADS 5). Small right pleural effusion.

VITAL SIGNS

Date	Time	Temp °C	HR	RR	BP	SpO ₂	O ₂ Delivery
01/08/24	14:32	38.9	104	26	148/88	88% RA	Room Air (arrival)
01/08/24	15:00	38.9	104	26	148/88	94%	4L NC
01/09/24	06:00	38.4	98	23	138/82	93%	4L NC
01/09/24	18:00	37.9	94	21	134/78	95%	3L NC
01/10/24	06:00	37.6	91	20	132/78	95%	3L NC
01/11/24	06:00	37.1	84	18	128/78	96%	2L NC
01/12/24	06:00	36.7	78	16	126/74	97%	1L NC
01/13/24	09:00	36.7	72	15	120/74	97% RA	Room Air

PROGRESS NOTES

01/09/2024 08:15 — Dr. Sarah K. Patel, MD

S: SpO₂ slightly improved. Cough productive. Appetite poor. Denies chest pain.
O: T 38.4°C · HR 98 · RR 23 · BP 138/82 · SpO₂ 93% on 4L NC · CRP 84.2 · WBC 11.8 · PCR POSITIVE confirmed
A/P: COVID-19 pneumonia with acute hypoxic respiratory failure. Continues 4L NC. Dexamethasone Day 2. Remdesivir Day 2. Pulmonology consulted. No escalation to HFNC at this time.

01/09/2024 14:30 — Dr. James R. Okonkwo, MD (Pulmonology Consult)

58M with confirmed COVID-19 pneumonia. P/F ratio 257 on room air. CT: bilateral GGOs 40–50% lung parenchyma, CO-RADS 5. Moderate-severe COVID-19 hypoxic respiratory failure.

Recommendations:

- Continue Dexamethasone 6 mg IV daily ×10 days total
- Complete Remdesivir 5-day course
- If O₂ requirement increases beyond 6L NC → HFNC 40L/min, FiO₂ 0.5
- Awake prone positioning minimum 2h TID

- Tocilizumab not indicated (CRP <150, not meeting RECOVERY trial threshold)
- Monitor for secondary bacterial superinfection (PCT 0.18 — low risk)

01/10/2024 08:45 — Dr. Sarah K. Patel, MD

S: Significant improvement. Speaking in full sentences. Appetite returning.
O: T 37.6°C · HR 91 · RR 20 · SpO₂ 95% on 3L NC · CRP 61.4 ↓ · Blood Cx no growth ×48h
A/P: Improving COVID-19 pneumonia. O₂ down-titrated 4L → 3L NC. Dexamethasone Day 3. Remdesivir Day 3. Discharge planning initiated.

01/13/2024 09:45 — Dr. Sarah K. Patel, MD (Discharge Note)

O: T 36.7°C · HR 72 · RR 15 · BP 120/74 · SpO₂ 97–98% on Room Air
Patient ambulated in hallway ×2 without desaturation. All discharge criteria met. **DISCHARGE TO HOME.**

DISCHARGE SUMMARY

Date: 01/13/2024 **Author:** Dr. Sarah K. Patel, MD **e-Signature:** Verified — 01/13/2024 10:05

Principal Diagnosis	COVID-19 (U07.1) with Acute Hypoxic Respiratory Failure (J96.01)
Secondary Diagnoses	T2DM with Hyperglycemia (E11.65), Essential Hypertension (I10), Hyperlipidemia (E78.5)

Hospital Course

Mr. Mitchell was admitted January 8 with confirmed COVID-19 pneumonia. Presentation: SpO₂ 88% RA, RR 26, Temp 38.9°C, PCR positive at 19:22. CT chest: bilateral GGOs 40–50% lung parenchyma, CO-RADS 5. Treated with Dexamethasone 6 mg IV daily and Remdesivir ×5 days. Blood cultures: no growth. Procalcitonin normal. Inflammatory markers trended downward throughout stay. Patient weaned from 4L NC to room air by Day 5 (01/13). Discharge SpO₂ 97% RA. All criteria met for discharge January 13.

Discharge Medications

- Dexamethasone 4 mg PO daily ×3 days, then 2 mg PO daily ×2 days (taper)
- Metformin 1000 mg PO BID (resume)
- Lisinopril 10 mg PO daily
- Atorvastatin 40 mg PO nightly
- Jardiance 10 mg PO daily (resume)
- Enoxaparin 40 mg SQ daily ×7 days post-discharge (extended VTE prophylaxis)

Condition: Good **Disposition:** Home

Follow-up: PCP in 7 days · Pulmonology in 4 weeks · Return to ED if SpO₂ <92%, worsening dyspnea, or chest pain

CLAIM INFORMATION

Field	Value
Claim Number	CLM-2024-10471
Submitted	01/18/2024
Payer	BlueCross BlueShield Texas
DRG	178
Principal Dx	U07.1
Total Billed	\$24,850.00

Riverside Community Hospital · Houston, TX 77001

INPATIENT MEDICAL RECORD — CONFIDENTIAL

Patient: Vasquez, Eleanor M. **MRN:** MRN-2024-002291 **DOB:** 09/22/1956 **Age:** 67 **Sex:** Female
Insurance: Aetna Medicare Advantage **Subscriber ID:** AET-MA-9934812
Admission: 03/14/2024 11:50 **Discharge:** 03/20/2024 13:30 **LOS:** 6 Days
Attending: Dr. Maria L. Santos, MD (Hospitalist)
Consultants: Dr. Kwame Asante, MD (Infectious Disease) · Dr. Rachel Kim, MD (Nephrology)

Admitting Diagnosis: Urinary Tract Infection with Systemic Sepsis

Discharge Diagnoses

	ICD-10	Description
Principal	A41.51	Sepsis due to Escherichia coli
Secondary	N39.0	Urinary Tract Infection (source)
Secondary	N17.9	Acute Kidney Injury
Secondary	J96.00	Acute Respiratory Failure
Secondary	F05	Delirium due to physiological condition
Secondary	E11.9	Type 2 Diabetes Mellitus
Secondary	I10	Essential Hypertension

	ICD-10	Description
Secondary	N18.3	Chronic Kidney Disease, Stage 3

DRG: 871 — Septicemia/Severe Sepsis without MV ≥96h with MCC **Billed:** \$38,200.00

ADMISSION HISTORY & PHYSICAL

Date: 03/14/2024 **Time:** 12:35 **Author:** Dr. Maria L. Santos, MD
(History obtained from daughter Maria Vasquez-Cruz — patient confused on arrival)

Chief Complaint

"She came in confused and has had a fever for 2 days."

History of Present Illness

Ms. Vasquez is a 67-year-old female with T2DM, Hypertension, and CKD Stage 3 presenting with 4 days of dysuria and urinary frequency, 2 days of fever (39.2°C at home), and 1 day of increasing confusion. Daughter reports patient not recognizing her at times. Decreased PO intake over the past 12 hours. Decreased urine output ×24 hours. No vomiting, diarrhea, or respiratory complaints reported by daughter.

Past Medical History

- Type 2 Diabetes Mellitus — Metformin 500 mg BID, Glipizide 5 mg daily
- Essential Hypertension — Amlodipine 5 mg, HCTZ 25 mg
- CKD Stage 3 — **Baseline creatinine 1.0 mg/dL** (per October 2023 records)
- Osteoporosis — Alendronate 70 mg weekly
- Hypothyroidism — Levothyroxine 75 mcg daily

Allergies: Penicillin (rash) · Sulfa (anaphylaxis)

Physical Examination

Parameter	Value	Interpretation
Temperature	39.4°C (103.0°F) rectal	Febrile
Heart Rate	118 bpm	Tachycardic
Respiratory Rate	22 breaths/min	Mildly elevated
Blood Pressure	88/52 mmHg	HYPOTENSIVE
SpO ₂	94% on Room Air	Low-normal
Weight	62 kg	
GCS	12 (E3V4M5)	Confused

- **Respiratory:** Clear to auscultation bilaterally. Non-labored breathing. RR 22.
- **CV:** Tachycardic. No murmurs. Cap refill 3 seconds.
- **Abdomen:** Soft. Mild suprapubic tenderness. No guarding.
- **Extremities:** Warm. No edema.
- **Neuro:** Confused, agitated. Not oriented to date or place.

Assessment & Plan

1. **SEPSIS — UTI source** — SEPSIS ALERT activated 12:05. Blood cultures ×2 obtained 12:30 (bilateral peripheral, before antibiotics). Urinalysis and urine culture ordered (catheter specimen). Lactate ordered (*order placed at 12:40*). NS 0.9% 30 mL/kg (~1,860 mL) bolus initiated 12:15. Ceftriaxone 2g IV — order placed 14:20, administered 14:47. Foley catheter for UO monitoring and specimen collection.
2. **AMS / Delirium** — Fall precautions. Reorientation.
3. **T2DM** — Hold Metformin (sepsis + CKD). Hold HCTZ (volume depleted). Glucose Q4h. Sliding scale.
4. **CKD Stage 3** — Monitor closely. Baseline Cr 1.0 mg/dL. Avoid nephrotoxins.

LABORATORY RESULTS

Initial Labs — 03/14/2024, 12:20

Complete Blood Count

Test	Result	Reference	Flag
WBC	18.4 K/uL	4.5–11.0	↑ HIGH
Neutrophils	91%	—	↑ HIGH
Bands	12%	0–5%	↑ HIGH
Hemoglobin	11.2 g/dL	12–16	↓ LOW
Platelets	142 K/uL	150–400	Low-normal

Comprehensive Metabolic Panel

Test	Result	Reference	Flag
Sodium	133 mEq/L	136–145	↓ Hyponatremia
BUN	42 mg/dL	7–20	↑ HIGH
Creatinine	1.6 mg/dL	Baseline: 1.0	↑ HIGH — AKI
eGFR	34 mL/min	Baseline: ~57	↓ Acute decline
Glucose	228 mg/dL	70–99	↑ HIGH
CO ₂	18 mEq/L	22–29	↓ Metabolic acidosis

Procalcitonin: 18.4 ng/mL ↑ (Reference <0.25) — markedly elevated

Urinalysis (catheter specimen, 12:45)

Parameter	Result
WBC	>100/hpf ↑
Bacteria	4+ (many)
Nitrites	POSITIVE
Leukocyte Esterase	POSITIVE (3+)

Lactate

Field	Value
Order Placed	12:40
Collection Time	16:30
Result	3.8 mmol/L (Reference 0.5–2.0) ↑ HIGH

Arterial Blood Gas — 03/14/2024 13:15

Parameter	Value	Reference	Flag
pH	7.29	7.35–7.45	↓ Acidosis
PaO ₂	78 mmHg	75–100	Low-normal
PaCO ₂	32 mmHg	35–45	↓ Compensation
HCO ₃	15 mEq/L	22–26	↓ LOW
SaO ₂	95% on Room Air	—	Adequate on RA

Blood Cultures ×2 (collected 12:30 — before antibiotics)

Timeline	Result
Preliminary 24h	Gram-negative rods in both bottles
Final 48h	<i>Escherichia coli</i> — pansensitive
Susceptibility	Ceftriaxone Ciprofloxacin Gentamicin

Urine Culture (12:45): *E. coli* >100,000 CFU/mL — matches blood isolate

Follow-up Labs

Date	WBC	Creatinine	Lactate	Notes
03/15	14.2 ↓	1.9 ↑ (peak)	1.8 (normalized)	
03/16	10.8 ↓	1.4 ↓	—	Improving

Date	WBC	Creatinine	Lactate	Notes
03/18	8.4 (normal)	1.2	—	Near baseline
03/19	7.8	1.1	—	Approaching baseline

RADIOLOGY

Chest X-Ray (PA) — 03/14/2024 13:00 (*Dr. Carlos Reyes, Radiology*)

Impression: Lungs clear bilaterally. No consolidation, effusion, or pneumothorax. Cardiac silhouette normal. **No acute cardiopulmonary process.**

VITAL SIGNS

Date	Time	Temp °C	HR	RR	BP	SpO ₂	UO mL/hr	O ₂
03/14	11:50	39.4	118	22	88/52	94% RA	—	Room Air
03/14	13:00	39.2	110	20	96/60	95% RA	8	Room Air
03/14	14:47	39.0	106	20	102/64	96% RA	20	Room Air (<i>ABX started</i>)
03/14	22:00	38.4	98	18	114/72	97% RA	45	Room Air
03/15	06:00	38.2	96	18	118/74	97% RA	52	Room Air
03/16	06:00	37.6	88	16	126/78	98% RA	62	Room Air
03/18	06:00	36.9	76	15	126/78	99% RA	72	Room Air
03/20	09:00	36.7	72	14	120/74	99% RA	66	Room Air

Note: Vasopressors were never administered. Patient maintained on Room Air throughout the entire 6-day hospitalization.

PROGRESS NOTES

03/14/2024 23:30 — Dr. Maria L. Santos, MD

S: Patient remains confused but less agitated. No respiratory complaints.

O: T 38.4°C · HR 98 · RR 18 · BP 114/72 · SpO₂ 97% RA · GCS improving · Blood Cx: GNRs both bottles (preliminary) · Lactate 3.8 mmol/L (collected 16:30) · Fluid balance: +1.4L

A/P: *E. coli* sepsis from UTI — continue Ceftriaxone 2g IV q24h; ID consult placed. AKI: Cr 1.6 from baseline 1.0 — continue IVF; Nephrology consult. Delirium: fall precautions; reorientation. No respiratory issues — SpO₂ 97% RA throughout shift.

03/15/2024 11:00 — Dr. Kwame Asante, MD (Infectious Disease Consult)

E. coli bacteremia from UTI source. Both blood cultures and urine culture confirm pansensitive *E. coli*. Susceptible to Ceftriaxone.

Recommendations:

- Continue Ceftriaxone 2g IV q24h (appropriate given allergy profile — PCN rash with low cephalosporin cross-reactivity)
- De-escalate to oral Ciprofloxacin 500 mg BID when tolerating PO to complete 14-day total course
- Repeat blood cultures in 48h to confirm clearance
- No echo or additional source workup needed — clear UTI source
- Initial MAP ~64 (BP 88/52): patient responded to IV fluids without vasopressors — vasopressor-free shock response

03/15/2024 14:00 — Dr. Rachel Kim, MD (Nephrology Consult)

AKI Stage 1 by KDIGO criteria — creatinine 1.6–1.9 vs baseline 1.0 (increase ≥ 0.3 mg/dL within 48h and $\geq 1.5\times$ baseline). Etiology: sepsis-associated prerenal/ATN. Recommend aggressive IV hydration, hold nephrotoxins, daily creatinine monitoring, adjust antibiotic dosing for renal function. No RRT indication at this time.

03/18/2024 08:30 — Dr. Maria L. Santos, MD

S: Patient alert and oriented $\times 3$ — back to baseline per daughter. Tolerating PO. Requesting to go home.

O: T 36.9°C · HR 76 · SpO₂ 99% RA · Cr 1.2 (near baseline 1.0) · WBC 8.4

A/P: Resolving *E. coli* sepsis. Tolerating PO. Transitioning to Ciprofloxacin 250 mg PO BID (renal-dose adjusted). Target discharge 03/20 after 2 days of observed oral antibiotics.

DISCHARGE SUMMARY

Date: 03/20/2024 **Author:** Dr. Maria L. Santos, MD **e-Signature:** Verified — 03/20/2024 12:45

Principal Diagnosis	Sepsis due to E. coli (A41.51)
Secondary Diagnoses	UTI (N39.0), AKI (N17.9), Acute Respiratory Failure (J96.00), Delirium (F05), T2DM (E11.9), HTN (I10), CKD Stage 3 (N18.3)

Hospital Course

Ms. Vasquez presented March 14 with urosepsis — E. coli bacteremia from UTI. BP 88/52, temp 39.4°C, altered mental status, initial lactate 3.8 mmol/L. Resuscitated with 30 mL/kg IV fluids — BP normalized without vasopressors. Ceftriaxone initiated and clinical course improved over 5 days. Delirium resolved with treatment of underlying infection. AKI improved with hydration. Transitioned to oral Ciprofloxacin; met discharge criteria on March 20.

Discharge Medications

- Ciprofloxacin 250 mg PO BID ×10 days
- Amlodipine 5 mg daily · Levothyroxine 75 mcg · Alendronate 70 mg weekly
- Metformin 500 mg BID (resume) · Glipizide 5 mg · HCTZ 25 mg (resume)

Condition: Good Disposition: Home with daughter

Follow-up: PCP in 1 week · Nephrology in 4 weeks

CLAIM INFORMATION

Field	Value
Claim Number	CLM-2024-23841
Submitted	03/25/2024
Payer	Aetna Medicare Advantage
DRG	871
Principal Dx	A41.51
Total Billed	\$38,200.00

Valley View Medical Center · Phoenix, AZ 85001

INPATIENT MEDICAL RECORD — CONFIDENTIAL

Patient: Chen, Harold K. MRN: MRN-2024-003742 DOB: 02/08/1952 Age: 72 Sex: Male
Insurance: Medicare Fee-for-Service ID: 1EG4-TE5-MK72

Admission: 04/02/2024 09:15 **Discharge:** 04/11/2024 14:00 **LOS:** 9 Days
Attending: Dr. Thomas A. Brennan, MD (Hospitalist)
Consultants: Dr. Yuen-Li Park, MD (Pulmonology) · Dr. Fatima Al-Rashidi, MD (Infectious Disease)

Admitting Diagnosis: COVID-19 Pneumonia with Acute Hypoxic Respiratory Failure

Discharge Diagnoses

	ICD-10	Description
Principal	A41.89	Sepsis due to other specified organism
Secondary	U07.1	COVID-19
Secondary	J18.9	Hospital-Acquired Pneumonia
Secondary	R65.21	Severe Sepsis with Septic Shock
Secondary	J96.01	Acute Respiratory Failure with Hypoxia
Secondary	E11.9	Type 2 Diabetes Mellitus
Secondary	N18.4	CKD Stage 4
Secondary	F05	Delirium

DRG: 871 — Septicemia/Severe Sepsis without MV ≥96h with MCC **Billed:** \$89,750.00

ADMISSION HISTORY & PHYSICAL

Date: 04/02/2024 **Time:** 10:45 **Author:** Dr. Thomas A. Brennan, MD

Chief Complaint

"Shortness of breath and fever for 6 days."

History of Present Illness

Mr. Chen is a 72-year-old male with Hypertension, T2DM, CKD Stage 4, and Atrial Fibrillation presenting with 6-day progressive dyspnea, fever, and cough. Home antigen test positive 5 days ago. Home SpO₂ 89–92% on personal pulse oximeter. Unable to complete sentences. Transported by ambulance.

Past Medical History

- Atrial Fibrillation — Apixaban 5 mg BID, Metoprolol 50 mg BID
- Hypertension — Amlodipine 10 mg, Lisinopril 20 mg
- Type 2 DM — Insulin Glargine 20 U QHS, Metformin 500 mg BID

4. **CKD Stage 4 — Baseline Creatinine ~2.8 mg/dL, eGFR ~18 mL/min/1.73m²**

5. GERD — Pantoprazole 40 mg daily

Social History: Widower. **Lives ALONE** in Phoenix, AZ. Retired engineer. Daughter lives in California.

Allergies: No Known Drug Allergies (NKDA)

Physical Examination

Parameter	Value	Interpretation
Temperature	38.7°C (101.7°F)	Febrile
Heart Rate	96 bpm (irregular — AF)	
Respiratory Rate	28 breaths/min	Elevated
Blood Pressure	136/82 mmHg	
SpO₂ (Room Air)	87%	Critically low
SpO ₂ after 4L NC	93%	
Weight	68 kg	

- **Respiratory:** Tachypneic. Accessory muscle use. Bilateral crackles. Decreased BS right base.
- **CV:** Irregularly irregular. No murmurs. 1+ bilateral ankle edema.

Assessment & Plan

1. **COVID-19 Pneumonia — Acute Hypoxic Respiratory Failure** — PCR ordered. 4L NC. Dexamethasone 6 mg IV daily. Remdesivir 200 mg IV Day 1 (*Note: patient CKD Stage 4 eGFR ~18 — ordered without documented renal consideration*). **HOLD Apixaban** (respiratory illness, fall risk — anticoagulation gap created). UFH 5000 U SQ q8h.
2. **AF** — Continue Metoprolol for rate control.
3. **CKD Stage 4** — Monitor closely. Adjust all medications for eGFR ~18.

LABORATORY RESULTS

Initial Labs — 04/02/2024

COVID-19 PCR: POSITIVE for SARS-CoV-2 RNA (resulted 04/02/2024 14:22)

CBC

Test	Result	Flag
WBC	13.4 K/uL	↑ HIGH
Lymphocytes	5%	↓ LOW
Hemoglobin	10.8 g/dL	↓ LOW (CKD anemia)

Test	Result	Flag
Platelets	164 K/uL	WNL

Inflammatory Markers

Test	Result	Flag
CRP	112 mg/L	↑ VERY HIGH
Ferritin	1,842 ng/mL	↑ VERY HIGH
D-Dimer	2.4 mcg/mL	↑ HIGH
Procalcitonin	0.31 ng/mL	Borderline ↑

Arterial Blood Gas — Room Air (04/02 10:00)

Parameter	Value	Flag
pH	7.36	WNL
PaO ₂	48 mmHg	↓ LOW — Hypoxemic respiratory failure
SaO ₂	83%	↓ LOW
P/F Ratio	229	Moderate hypoxemia

Day 4 Labs — 04/05/2024 (Day Sepsis Declared)

Test	Result	Flag
WBC	22.6 K/uL	↑ MARKED SPIKE — new bacterial infection
Bands	8%	↑ HIGH
CRP	218 mg/L	↑ MARKEDLY ELEVATED (from 112)
Procalcitonin	12.8 ng/mL	↑ HIGH — bacterial infection
Creatinine	3.8 mg/dL	↑ Worsening (from 3.1)

Blood Cultures — 04/05/2024

Field	Status
Order placed	14:20
Collection	Not documented in nursing records or lab system
Result	NOT RESULTED — no blood cultures in record

Lactate — 04/05/2024

Field	Status
Order placed	14:15
Specimen collected (per lab)	14:25
Result	NOT PRESENT IN EMR — result absent

Sputum Culture — 04/05/2024 15:00

Field	Value
Gram Stain	3+ Gram-positive cocci in clusters
Culture	<i>Staphylococcus aureus</i>
MRSA Screen	POSITIVE

VITAL SIGNS — Selected Days

Date	Time	Temp °C	HR	RR	BP	MAP	SpO ₂	O ₂	Note
04/02	09:15	38.7	96AF	28	136/82	99	87% RA	Room Air	Arrival
04/02	10:00	38.7	96AF	28	136/82	99	93%	4L NC	
04/03	06:00	38.4	102	22	128/76	93	93%	4L NC	<i>Nursing</i>
04/03	12:00	38.2	99	21	130/78	95	93%	4L NC	
04/05	14:00	38.9	104	24	124/76	92	90%	6L NC	Sepsis declared
04/05	18:00	39.1	108	25	122/74	90	89%	HFNC 40L/50 %	
04/05	22:00	38.8	102	22	126/78	94	92%	HFNC	

Vasopressors administered: None throughout entire hospitalization.

PROGRESS NOTES

04/03/2024 09:00 — Dr. Thomas A. Brennan, MD

O: T 38.2°C · **HR 88** · **RR 18** · **BP 140/84** · SpO₂ 93% on 4L NC

(*Nursing flow sheet 06:00 same date records: HR 102 · RR 22 · BP 128/76*)

A/P: COVID-19 pneumonia. Remdesivir Day 2. Dexamethasone Day 2. Patient appears improved

from admission. Breathing more comfortably.

04/05/2024 14:30 — Dr. Thomas A. Brennan, MD

O: T 38.9°C · HR 104 · RR 24 · BP 124/76 (MAP 92) · SpO₂ 90% on 6L NC

A/P: Patient has developed hospital-acquired sepsis, likely MRSA pneumonia superimposed on COVID-19. Starting broad-spectrum antibiotics. Patient is in septic shock — will closely monitor hemodynamics. Blood cultures ×2 ordered. Lactate ordered. Start Vancomycin 2000 mg IV q12h + Piperacillin-Tazobactam 3.375 g IV q8h.

Medication Administration Record — Vancomycin

Field	Value
Order	Vancomycin 2000 mg IV q12h
Patient Weight	68 kg
Standard 15–20 mg/kg	1,020–1,360 mg/dose
Ordered dose	2,000 mg = 29.4 mg/kg
First dose administered	04/05/2024 19:30
Pharmacy consult	Not documented
Trough monitoring	Not ordered

Billing Records

Date	CPT	Description
04/05/2024	99291	Critical Care, first 30–74 min
04/06/2024	99291	Critical Care, first 30–74 min

Patient location per all nursing documentation: Room 412, Medical-Surgical Unit throughout entire hospitalization. No ICU transfer order, no ICU nursing notes, no ICU flowsheets present in chart.

04/09/2024 13:00 — Sarah Whitmore, MSW (Social Work)

Patient assessed for discharge planning. 72-year-old widower — **lives ALONE** in Phoenix. Daughter in California. Patient preference: **discharge to HOME**. Patient explicitly declined SNF. Daughter agrees patient should go home. Home health referral placed. **Plan: Discharge to HOME with home health aide 3×/week.**

DISCHARGE SUMMARY

Date: 04/11/2024

Author: Dr. Thomas A. Brennan, MD

Signature: *Verified by: Michael Chen, MS-4 (Medical Student)* — No attending physician electronic signature present

Principal Diagnosis	Sepsis (A41.89)
Secondary Diagnoses	COVID-19 (U07.1), HAP (J18.9), Septic Shock (R65.21), ARF (J96.01), T2DM (E11.9), HTN (I10), CKD Stage 4 (N18.4)

Hospital Course

Mr. Chen was admitted April 2 for COVID-19 pneumonia with hypoxic respiratory failure. Treated with Dexamethasone and Remdesivir. On Day 4, developed hospital-acquired pneumonia with septic shock requiring broad-spectrum antibiotics including Vancomycin and Pip-Tazo for MRSA coverage. He was transferred to the ICU for critical care management and stabilized over the following days. He was subsequently weaned from HFNC and transferred to the medical floor prior to discharge.

Discharge Medications

- Dexamethasone PO taper · Ciprofloxacin 500 mg PO BID ×7 days · Metoprolol 50 mg BID
- Amlodipine 10 mg · Lisinopril 20 mg · Apixaban 5 mg BID (resume) · Insulin Glargine 20 U QHS · Pantoprazole 40 mg

Condition: Fair **Disposition:** Skilled Nursing Facility

CLAIM INFORMATION

Field	Value
Claim Number	CLM-2024-47831
Submitted	04/18/2024
Payer	Medicare FFS
DRG	871
Principal Dx	A41.89
Total Billed	\$89,750.00